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CKAD Simulator Kubernetes 1.33

https://killer.sh

Each question needs to be solved on a specific instance other than your main `candidate@terminal`. You'll need to connect to the correct instance via ssh, the command is provided before each question. To connect to a different instance you always need to return first to your main terminal by running the `exit` command, from there you can connect to a different one.

In the real exam each question will be solved on a different instance whereas in the simulator multiple questions will be solved on same instances.

Question 1 | Namespaces

Solve this question on instance: `ssh ckad5601`

The DevOps team would like to get the list of all *Namespaces* in the cluster.

The list can contain other columns like STATUS or AGE.

Save the list to `/opt/course/1/namespaces` on `ckad5601`.

Answer:

```
k get ns > /opt/course/1/namespaces
```

The content should then look like:

```
# /opt/course/1/namespaces
NAME              STATUS   AGE
default           Active   136m
earth             Active   105m
jupiter           Active   105m
kube-node-lease   Active   136m
kube-public       Active   136m
kube-system       Active   136m
mars              Active   105m
shell-intern      Active   105m
```

Question 2 | Pods

Solve this question on instance: `ssh ckad5601`

Create a single *Pod* of image `httpd:2.4.41-alpine` in *Namespace* `default`. The *Pod* should be named `pod1` and the container should be named `pod1-container`.

Your manager would like to run a command manually on occasion to output the status of that exact *Pod*. Please write a command that does this into `/opt/course/2/pod1-status-command.sh` on `ckad5601`. The command should use `kubectl`.

Answer:

```
k run # help

k run pod1 --image=httpd:2.4.41-alpine --dry-run=client -oyaml > 2.yaml

vim 2.yaml
```

Change the container name in `2.yaml` to `pod1-container`:

```
# 2.yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: pod1
  name: pod1
spec:
  containers:
  - image: httpd:2.4.41-alpine
    name: pod1-container # change
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
```

Then run:

```
➜ k create -f 2.yaml
pod/pod1 created

➜ k get pod
NAME   READY   STATUS              RESTARTS   AGE
pod1   0/1     ContainerCreating   0          6s

➜ k get pod
NAME   READY   STATUS    RESTARTS   AGE
pod1   1/1     Running   0          30s
```

Next create the requested command:

```
vim /opt/course/2/pod1-status-command.sh
```

The content of the command file could look like:

```
# /opt/course/2/pod1-status-command.sh
kubectl -n default describe pod pod1 | grep -i status:
```

Another solution would be using jsonpath:

```
# /opt/course/2/pod1-status-command.sh
kubectl -n default get pod pod1 -o jsonpath="{.status.phase}"
```

To test the command:

```
➜ sh /opt/course/2/pod1-status-command.sh
Running
```

Question 3 | Job

Solve this question on instance: `ssh ckad7326`

Team Neptune needs a *Job* template located at `/opt/course/3/job.yaml`. This *Job* should run image `busybox:1.31.0` and execute `sleep 2 && echo done`. It should be in namespace `neptune`, run a total of 3 times and should execute 2 runs in parallel.

Start the *Job* and check its history. Each pod created by the *Job* should have the label `id: awesome-job`. The job should be named `neb-new-job` and the container `neb-new-job-container`.

Answer:

```
k -n neptune create job -h

k -n neptune create job neb-new-job --image=busybox:1.31.0 --dry-run=client -oyaml -- sh -c "sleep 2 && echo done" >
/opt/course/3/job.yaml

vim /opt/course/3/job.yaml
```

Make the required changes in the yaml:

```
# /opt/course/3/job.yaml
apiVersion: batch/v1
kind: Job
metadata:
  creationTimestamp: null
  name: neb-new-job
  namespace: neptune
spec:
  completions: 3          # add
  parallelism: 2          # add
  template:
    metadata:
      creationTimestamp: null
      labels:             # add
        id: awesome-job   # add
    spec:
      containers:
      - command:
        - sh
        - -c
        - sleep 2 && echo done
        image: busybox:1.31.0
        name: neb-new-job-container # update
        resources: {}
      restartPolicy: Never
status: {}
```

Then to create it:

```
k -f /opt/course/3/job.yaml create # namespace already set in yaml hence not needed
```

Check *Job* and *Pods*, you should see two running parallel at most but three in total:

```
➜ k -n neptune get pod,job | grep neb-new-job
pod/neb-new-job-jhq2g              0/1     ContainerCreating   0          4s
pod/neb-new-job-vf6ts              0/1     ContainerCreating   0          4s

job.batch/neb-new-job   0/3           4s         5s


➜ k -n neptune get pod,job | grep neb-new-job
pod/neb-new-job-gm8sz              0/1     ContainerCreating   0          0s
pod/neb-new-job-jhq2g              0/1     Completed           0          10s
pod/neb-new-job-vf6ts              1/1     Running             0          10s

job.batch/neb-new-job   1/3           10s        11s


➜ k -n neptune get pod,job | grep neb-new-job
pod/neb-new-job-gm8sz              0/1     ContainerCreating   0          5s

pod/neb-new-job-jhq2g              0/1     Completed           0          15s
pod/neb-new-job-vf6ts              0/1     Completed           0          15s
job.batch/neb-new-job   2/3           15s        16s


➜ k -n neptune get pod,job | grep neb-new-job
pod/neb-new-job-gm8sz              0/1     Completed          0          12s
pod/neb-new-job-jhq2g              0/1     Completed          0          22s
pod/neb-new-job-vf6ts              0/1     Completed          0          22s

job.batch/neb-new-job   3/3           21s        23s
```

Check history:

```
➜ k -n neptune describe job neb-new-job
...
Events:
  Type    Reason            Age    From            Message
  ----    ------            ----   ----            -------
  Normal  SuccessfulCreate  2m52s  job-controller  Created pod: neb-new-job-jhq2g
  Normal  SuccessfulCreate  2m52s  job-controller  Created pod: neb-new-job-vf6ts
  Normal  SuccessfulCreate  2m42s  job-controller  Created pod: neb-new-job-gm8sz
```

At the age column we can see that two `pods` run parallel and the third one after that. Just as it was required in the task.

Question 4 | Helm Management

Solve this question on instance: `ssh ckad7326`

Team Mercury asked you to perform some operations using Helm, all in *Namespace* `mercury`:

1. Delete release `internal-issue-report-apiv1`
2. Upgrade release `internal-issue-report-apiv2` to any newer version of chart `bitnami/nginx` available
3. Install a new release `internal-issue-report-apache` of chart `bitnami/apache`. The *Deployment* should have two replicas, set these via Helm-values during install
4. There seems to be a broken release, stuck in `pending-install` state. Find it and delete it

Answer:

Helm Chart: Kubernetes YAML template-files combined into a single package, *Values* allow customisation

Helm Release: Installed instance of a *Chart*

Helm Values: Allow to customise the YAML template-files in a *Chart* when creating a *Release*

Step 1

First we should delete the required release:

```
➜ helm -n mercury ls
NAME                            NAMESPACE     ...   STATUS      CHART           APP VERSION
internal-issue-report-apiv1     mercury       ...   deployed    nginx-18.1.14   1.27.1
internal-issue-report-apiv2     mercury       ...   deployed    nginx-18.1.14   1.27.1
internal-issue-report-app       mercury       ...   deployed    nginx-18.1.14   1.27.1

➜ helm -n mercury uninstall internal-issue-report-apiv1
release "internal-issue-report-apiv1" uninstalled

➜ helm -n mercury ls
NAME                            NAMESPACE     ...   STATUS      CHART           APP VERSION
internal-issue-report-apiv2     mercury       ...   deployed    nginx-18.1.14   1.27.1
internal-issue-report-app       mercury       ...   deployed    nginx-18.1.14   1.27.1
```

Step 2

Next we need to upgrade a release, for this we could first list the charts of the repo:

```
➜ helm repo list
NAME    URL
bitnami http://localhost:6000

➜ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "bitnami" chart repository
Update Complete. ⎈Happy Helming!⎈

➜ helm search repo nginx --versions
NAME            CHART VERSION   APP VERSION     DESCRIPTION
bitnami/nginx   18.2.0          1.27.1          NGINX Open Source is a web server that can be a...
bitnami/nginx   18.1.15         1.27.1          NGINX Open Source is a web server that can be a...
bitnami/nginx   18.1.14         1.27.1          NGINX Open Source is a web server that can be a...
bitnami/nginx   13.0.0          1.23.0          NGINX Open Source is a web server that can be a...
```

Here we see that two newer chart versions are available. But the question only requires us to upgrade to any newer chart version available, so we can simply run:

```
➜ helm -n mercury upgrade internal-issue-report-apiv2 bitnami/nginx
Release "internal-issue-report-apiv2" has been upgraded. Happy Helming!
NAME: internal-issue-report-apiv2
```

```
LAST DEPLOYED: Wed Oct  2 14:17:09 2024
NAMESPACE: mercury
STATUS: deployed
REVISION: 2
TEST SUITE: None
NOTES:
CHART NAME: nginx
CHART VERSION: 18.2.0
APP VERSION: 1.27.1
...

➜ helm -n mercury ls
NAME                            NAMESPACE   ...   STATUS      CHART           APP VERSION
internal-issue-report-apiv2     mercury     ...   deployed    nginx-18.2.0    1.27.1
internal-issue-report-app       mercury     ...   deployed    nginx-18.1.14   1.27.1
```

Looking good!

> ***INFO:*** Also check out `helm rollback` for undoing a helm rollout/upgrade

Step 3

Now we're asked to install a new release, with a customised values setting. For this we first list all possible value settings for the chart, we can do this via:

```
helm show values bitnami/apache # will show a long list of all possible value-settings

helm show values bitnami/apache | yq e # parse yaml and show with colors
```

Huge list, if we search in it we should find the setting `replicaCount: 1` on top level. This means we can run:

```
➜ helm -n mercury install internal-issue-report-apache bitnami/apache --set replicaCount=2
NAME: internal-issue-report-apache
LAST DEPLOYED: Wed Oct  2 14:18:32 2024
NAMESPACE: mercury
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
CHART NAME: apache
CHART VERSION: 11.2.20
APP VERSION: 2.4.62
...
```

If we would also need to set a value on a deeper level, for example `image.debug`, we could run:

```
helm -n mercury install internal-issue-report-apache bitnami/apache \
  --set replicaCount=2 \
  --set image.debug=true
```

Install done, let's verify what we did:

```
➜ helm -n mercury ls
NAME                            NAMESPACE     ...   STATUS       CHART              APP VERSION
internal-issue-report-apache    mercury       ...   deployed     apache-11.2.20     2.4.62
internal-issue-report-apiv2     mercury       ...   deployed     nginx-18.2.0       1.27.1
internal-issue-report-app       mercury       ...   deployed     nginx-18.1.14      1.27.1

➜ k -n mercury get deploy internal-issue-report-apache
NAME                           READY   UP-TO-DATE   AVAILABLE   AGE
internal-issue-report-apache   2/2     2            2           64s
```

We see a healthy deployment with two replicas!

Step 4

By default releases in `pending-upgrade` state aren't listed, but we can show all to find and delete the broken release:

```
➜ helm -n mercury ls -a
NAME                            NAMESPACE   ...   STATUS           CHART              APP VERSION
internal-issue-report-apache    mercury     ...   deployed         apache-11.2.20     2.4.62
internal-issue-report-apiv2     mercury     ...   deployed         nginx-18.2.0       1.27.1
internal-issue-report-app       mercury     ...   deployed         nginx-18.1.14      1.27.1
internal-issue-report-daniel    mercury     ...   pending-install  nginx-18.1.14      1.27.1

➜ helm -n mercury uninstall internal-issue-report-daniel
release "internal-issue-report-daniel" uninstalled
```

Thank you Helm for making our lives easier! (Till something breaks)

Question 5 | ServiceAccount, Secret

Solve this question on instance: `ssh ckad7326`

Team Neptune has its own *ServiceAccount* named `neptune-sa-v2` in *Namespace* `neptune`. A coworker needs the token from the *Secret* that belongs to that *ServiceAccount*. Write the base64 decoded token to file `/opt/course/5/token` on `ckad7326`.

Answer:

Secrets won't be created automatically for *ServiceAccounts, but it's possible to create a *Secret* manually and attach it to a *ServiceAccount* by setting the correct annotation on the *Secret*. This was done for this task.

```
k -n neptune get sa # get overview
k -n neptune get secrets # shows all secrets of namespace
k -n neptune get secrets -oyaml | grep annotations -A 1 # shows secrets with first annotation
```

If a *Secret* belongs to a *ServiceAccount*, it'll have the annotation `kubernetes.io/service-account.name`. Here the *Secret* we're looking for is `neptune-secret-1`.

```
➜ k -n neptune get secret neptune-secret-1 -o yaml
apiVersion: v1
data:
...
  token:
ZXlKaGJHY2lPaUpTVXpJMU5pSXNJbXRwWkNJNkltNWFaRmRxWkRKMmFHTnZRM0JxV0haT1IxZzFiM3BJY2O1SlowaEhOV3hUWmt3elFuRmFhVEZhZDJNaW
ZRLmV5SnBjM01pT2lKcmRXSmxjbTVsZEdWekwzTmxjblpwWTJWaFkyTnZkVzUwSWl3aWEzVmlaWEp1WlhSbGN5NXBieTl6WlhKMmFXTmxZV05qYjNWdWRD
OXVZVzFsYzNCaFkyVWlPaUp1WlhCMGRXNWxJaXdpYTNWaVpYSnVaWFJsY3k1cGJ5OXpaWEoyYVdObFlXTmpiM1Z1ZEM5elpXTnlaWFF1Ym1GdFpTSTZJbT
VsY0hSMWJtVXRjMkV0ZGpJdGRHOXJaVzR0Wm5FNU1tb2lMQ0pyZFdKbGNtNWxkR1Z6TG1sdkwzTmxjblpwWTJWaFkyTnZkVzUwTDNObGNuWnBZMlV0WVdO
amIzVnVkQzV1WVcxbElqb2libVZ3ZEhWdVpTMXpZUzEyTWlJc0ltdDFZbVZ5Ym1WMFpYTXVhVzh2YzJWeWRtbGpaV0ZqWTI5MWJuUXZjMlZ5ZG1salpTMW
hZMk52ZFc1MExuVnBaQ0k2SWpZMlltUmpOak0yTFRKbFl6TXROREpoWkMwNE9HRTFMV0ZoWXpGbFpqWmxPVFpsTlNJc0luTjFZaUk2SW5ONWMzUmxiVHB6
WlhKMmFXTmxZV05qYjNWdWREcHVaWEIwZFc1bE9tNWxjSFIxYm1VdGMyRXRkaklpZlEuVllnYm9NNENUZDBwZENKNzh3alV3bXRhbGgtMnZzS2pBTnlQc2
gtNmd1RXdPdFdFcTVGYnc1WkhQdHZBZHJMbFB6cE9IRWJBZTRlVU5NUJSR1diWUlkd2p1Tjk1SjBENFJORmtWVXQ0OHR3b2FrUlY3aC1hUHV3c1FYSGha
Wnp5NHlpbUZIRzlVZm1zazVZcjRSVmNHNm4xMzd5LUZIMDhLOHpaaklQQXNLRHFOQlFOeGctbFp2d1ZNaTZ2aUlocnJ6QVFzME1CT1Y4Mk9KWUd5Mm8tV1
FWYzBVVWFuQ2Y5NFkzZ1QwWVRpcVF2Y3pZTXM2bno5dXQtWGd3aXRyQlk2VGo5QmdQcHJBOWtfajVxRXhfTFVVWlVwUEFpRU43T3pka0pzSThjdHRoMTBs
eXBJMUFlRnIOM3Q2QUx5clFvQk0zOWFiRGZxM0Zrc1Itb2NfV013
kind: Secret
...
```

This shows the base64 encoded token. To get the decoded one we could pipe it manually through `base64 -d` or we simply do:

```
➜ k -n neptune describe secret neptune-secret-1
...
Data
====
token:
eyJhbGciOiJSUzI1NiIsImtpZCI6Im5aZFdqZDJ2aGNvQ3BqWHZOR1g1b3pIcm5JZ0hHNWxTZkwzQnFaaTFad2MifQ.eyJpc3MiOiJrdWJlcm5ldGVzL3N
lcnZpY2VhY2NvdW50Iiwia3ViZXJuZXRlcy5pby9zZXJ2aWNlYWNjb3VudC9uYW1lc3BhY2UiOiJuZXB0dW5lIiwia3ViZXJuZXRlcy5pby9zZXJ2aWNlY
WNjb3VudC9zZWNyZXQubmFtZSI6Im5lcHR1bmUtc2EtdjItdG9rZW4tZnE5MmoiLCJrdWJlcm5ldGVzLmlvL3NlcnZpY2VhY2NvdW50L3NlcnZpY2UtYWN
jb3VudC5uYW1lIjoibmVwdHVuZS1zYS12MiIsImt1YmVybmV0ZXMuaW8vc2VydmljZWFjY291bnQvc2VydmljZS1hY2NvdW50LnVpZCI6IjY2YmRjNjM2L
TJlYzMtNDJhZC04OGE1LWFhYzFlZjZlOTZlNSIsInN1YiI6InN5c3RlbTpzZXJ2aWNlYWNjb3VudDpuZXB0dW5lOm5lcHR1bmUtc2EtdjIifQ.vYgboM4C
Td0pdCJ78wjUwmtalh-2vsKjANyPsh-6guEwOtWEq5Fbw5ZHPtvAdrLlPzpOHEbAe4eUM95BRGWbYIdwjuN95JOD4RNFkVUt48twoakRV7h-
aPuwsQXHhZZzy4yimFHG9Ufmsk5Yr4RVcG6n137y-FHO8K8zZjIPAsKDqNBQtxg-lZvwVMi6viIhrrzAQsOMBOV82OJYGy2o-
WQVcOUUanCf94Y3gTOYTiqQvczYMs6nz9ut-
XgwitrBY6Tj9BgPprA9k_j5qEx_LUUZUpPAiEN7OzdkJsI8ctth10lypI1AeFr43t6ALyrQoBM39abDfq3FksR-oc_WMw
ca.crt:     1066 bytes
namespace:  7 bytes
```

Copy the token (part under `token:`) and paste it using vim.

```
vim /opt/course/5/token
```

File `/opt/course/5/token` should contain the token:

```
# /opt/course/5/token
eyJhbGciOiJSUzI1NiIsImtpZCI6Im5aZFdqZDJ2aGNvQ3BqWHZOR1g1b3pIcm5JZOhHNWxTZkwzQnFaaTFad2MifQ.eyJpc3MiOiJrdWJlcm5ldGVzL3NlcnZpY2VhY2NvdW5OIiwia3ViZXJuZXRlcy5pby9zZXJ2aWNlYWNjb3VudC9uYW1lc3BhY2UiOiJuZXBOdW5lIiwia3ViZXJuZXRlcy5pby9zZXJ2aWNlYWNjb3VudC9zZWNyZXQubmFtZSI6Im5lcHR1bmUtc2EtdjItdG9rZW4tZnE5MmoiLCJrdWJlcm5ldGVzLmlvL3NlcnZpY2VhY2NvdW5OL3NlcnZpY2UtYWNjb3VudC5uYW1lIjoibmVwdHVuZS1zYS12MiIsImt1YmVybmV0ZXMuaW8vc2VydmljZWFjY291bnQvc2VydmljZS1hY2NvdW5OLnVpZCI6IjY2YmRjNjM2LTJlYzMtNDJhZC04OGE1LWFhYzFlZjZlOTZlNSIsInN1YiI6InN5c3RlbTpzZXJ2aWNlYWNjb3VudDpuZXBOdW5lOm5lcHR1bmUtc2EtdjIifQ.VYgboM4CTd0pdCJ78wjUwmtalh-2vsKjANyPsh-6guEwOtWEq5Fbw5ZHPtvAdrLlPzpOHEbAe4eUM95BRGWbYIdwjuN95JOD4RNFkVUt48twoakRV7h-aPuwsQXHhZZzy4yimFHG9Ufmsk5Yr4RVcG6n137y-FHO8K8zZjIPAsKDqNBQtxg-lZvwVMi6viIhrrzAQs0MBOV82OJYGy2o-WQVcOUUanCf94Y3gTOYTiqQvczYMs6nz9ut-XgwitrBY6Tj9BgPprA9k_j5qEx_LUUZUpPAiEN7OzdkJsI8ctth10lypI1AeFr43t6ALyrQoBM39abDfq3FksR-oc_WMw
```

Question 6 | ReadinessProbe

Solve this question on instance: `ssh ckad5601`

Create a single *Pod* named `pod6` in *Namespace* `default` of image `busybox:1.31.0`. The *Pod* should have a readiness-probe executing `cat /tmp/ready`. It should initially wait 5 and periodically wait 10 seconds. This will set the container ready only if the file `/tmp/ready` exists.

The *Pod* should run the command `touch /tmp/ready && sleep 1d`, which will create the necessary file to be ready and then idles. Create the *Pod* and confirm it starts.

Answer:

```
k run pod6 --image=busybox:1.31.0 --dry-run=client -oyaml --command -- sh -c "touch /tmp/ready && sleep 1d" > 6.yaml

vim 6.yaml
```

Search for a readiness-probe example on https://kubernetes.io/docs, then copy and alter the relevant section for the task:

```
# 6.yaml
apiVersion: v1
kind: Pod
```

```
metadata:
  creationTimestamp: null
  labels:
    run: pod6
  name: pod6
spec:
  containers:
  - command:
    - sh
    - -c
    - touch /tmp/ready && sleep 1d
    image: busybox:1.31.0
    name: pod6
    resources: {}
    readinessProbe:                          # add
      exec:                                  # add
        command:                             # add
        - sh                                 # add
        - -c                                 # add
        - cat /tmp/ready                     # add
      initialDelaySeconds: 5                 # add
      periodSeconds: 10                      # add
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
```

Then:

```
k -f 6.yaml create
```

Running `k get pod6` we should see the job being created and completed:

```
➜ k get pod pod6
NAME   READY   STATUS              RESTARTS   AGE
pod6   0/1     ContainerCreating   0          2s

➜ k get pod pod6
NAME   READY   STATUS    RESTARTS   AGE
pod6   0/1     Running   0          7s

➜ k get pod pod6
NAME   READY   STATUS    RESTARTS   AGE
pod6   1/1     Running   0          15s
```

We see that the *Pod* is finally ready.

Question 7 | Pods, Namespaces

Solve this question on instance: `ssh ckad7326`

The board of Team Neptune decided to take over control of one e-commerce webserver from Team Saturn. The administrator who once setup this webserver is not part of the organisation any longer. All information you could get was that the e-commerce system is called `my-happy-shop`.

Search for the correct *Pod* in *Namespace* `saturn` and move it to *Namespace* `neptune`. It doesn't matter if you shut it down and spin it up again, it probably hasn't any customers anyways.

Answer:

Let's see all those *Pods*:

```
➜ k -n saturn get pod
NAME                READY   STATUS    RESTARTS   AGE
webserver-sat-001   1/1     Running   0          111m
webserver-sat-002   1/1     Running   0          111m
webserver-sat-003   1/1     Running   0          111m
webserver-sat-004   1/1     Running   0          111m
webserver-sat-005   1/1     Running   0          111m
webserver-sat-006   1/1     Running   0          111m
```

The *Pod* names don't reveal any information. We assume the *Pod* we are searching has a *label* or *annotation* with the name `my-happy-shop`, so we search for it:

```
k -n saturn describe pod # describe all pods, then manually look for it

# or do some filtering like this
k -n saturn get pod -o yaml | grep my-happy-shop -A10
```

We see the webserver we're looking for is `webserver-sat-003`

```
k -n saturn get pod webserver-sat-003 -o yaml > 7_webserver-sat-003.yaml # export
vim 7_webserver-sat-003.yaml
```

Change the *Namespace* to `neptune`, also remove the `status:` section, the token `volume`, the token `volumeMount` and the `nodeName`, else the new *Pod* won't start. The final file could look as clean like this:

```
# 7_webserver-sat-003.yaml
apiVersion: v1
kind: Pod
metadata:
  annotations:
    description: this is the server for the E-Commerce System my-happy-shop
  labels:
    id: webserver-sat-003
  name: webserver-sat-003
  namespace: neptune # new namespace here
spec:
  containers:
  - image: nginx:1.16.1-alpine
    imagePullPolicy: IfNotPresent
    name: webserver-sat
  restartPolicy: Always
```

Then we execute:

```
k -n neptune create -f 7_webserver-sat-003.yaml
```

```
➜ k -n neptune get pod | grep webserver
webserver-sat-003               1/1     Running            0          22s
```

It seems the server is running in *Namespace* `neptune`, so we can do:

```
k -n saturn delete pod webserver-sat-003 --force --grace-period=0
```

Let's confirm only one is running:

```
➜ k get pod -A | grep webserver-sat-003
neptune        webserver-sat-003         1/1     Running            0          6s
```

This should list only one pod called `webserver-sat-003` in *Namespace* `neptune`, status running.

Question 8 | Deployment, Rollouts

Solve this question on instance: `ssh ckad7326`

There is an existing *Deployment* named `api-new-c32` in *Namespace* `neptune`. A developer did make an update to the *Deployment* but the updated version never came online. Check the *Deployment* history and find a revision that works, then rollback to it. Could you tell Team Neptune what the error was so it doesn't happen again?

Answer:

```
k -n neptune get deploy # overview
k -n neptune rollout -h
k -n neptune rollout history -h
```

```
➜ k -n neptune rollout history deploy api-new-c32
deployment.extensions/api-new-c32
REVISION  CHANGE-CAUSE
1         <none>
2         kubectl edit deployment api-new-c32 --namespace=neptune
3         kubectl edit deployment api-new-c32 --namespace=neptune
4         kubectl edit deployment api-new-c32 --namespace=neptune
5         kubectl edit deployment api-new-c32 --namespace=neptune
```

We see 5 revisions, let's check *Pod* and *Deployment* status:

```
➜ k -n neptune get deploy,pod | grep api-new-c32
deployment.extensions/api-new-c32    3/3     1            3           141m

pod/api-new-c32-65d998785d-jtmqq    1/1     Running            0          141m
pod/api-new-c32-686d6f6b65-mj2fp    1/1     Running            0          141m
pod/api-new-c32-6dd45bdb68-2p462    1/1     Running            0          141m
pod/api-new-c32-7d64747c87-zh648    0/1     ImagePullBackOff   0          141m
```

Let's check the pod for errors:

```
➜ k -n neptune describe pod api-new-c32-7d64747c87-zh648 | grep -i error
  ...  Error: ImagePullBackOff
```

```
➜ k -n neptune describe pod api-new-c32-7d64747c87-zh648 | grep -i image
    Image:          ngnix:1.16.3
    Image ID:
      Reason:       ImagePullBackOff
  Warning  Failed  4m28s (x616 over 144m)  kubelet, gke-s3ef67020-28c5-45f7--default-pool-248abd4f-s010  Error:
ImagePullBackOff
```

Someone seems to have added a new image with a spelling mistake in the name `ngnix:1.16.3`, that's the reason we can tell Team Neptune!

Now let's revert to the previous version:

```
k -n neptune rollout undo deploy api-new-c32
```

Does this one work?

```
➜ k -n neptune get deploy api-new-c32
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
api-new-c32   3/3     3            3           146m
```

Yes! All up-to-date and available.

Also a fast way to get an overview of the *ReplicaSets* of a *Deployment* and their images could be done with:

```
k -n neptune get rs -o wide | grep api-new-c32
```

Question 9 | Pod -> Deployment

Solve this question on instance: `ssh ckad9043`

In *Namespace* `pluto` there is single *Pod* named `holy-api`. It has been working okay for a while now but Team Pluto needs it to be more reliable.

Convert the *Pod* into a *Deployment* named `holy-api` with 3 replicas and delete the single *Pod* once done. The raw *Pod* template file is available at `/opt/course/9/holy-api-pod.yaml`.

In addition, the new *Deployment* should set `allowPrivilegeEscalation: false` and `privileged: false` for the security context on container level.

Please create the *Deployment* and save its yaml under `/opt/course/9/holy-api-deployment.yaml` on `ckad9043`.

Answer

There are multiple ways to do this, one is to copy an *Deployment* example from https://kubernetes.io/docs and then merge it with the existing *Pod* yaml. That's what we will do now:

```
cp /opt/course/9/holy-api-pod.yaml /opt/course/9/holy-api-deployment.yaml # make a copy!

vim /opt/course/9/holy-api-deployment.yaml
```

Now copy/use a *Deployment* example yaml and put the *Pod's* **metadata:** and **spec:** into the *Deployment's* **template:** section:

```
# /opt/course/9/holy-api-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: holy-api        # name stays the same
  namespace: pluto      # important
spec:
  replicas: 3           # 3 replicas
  selector:
    matchLabels:
      id: holy-api      # set the correct selector
```

```
  template:
    # => from here down it's the same as the pods metadata: and spec: sections
    metadata:
      labels:
        id: holy-api
      name: holy-api
    spec:
      containers:
      - env:
        - name: CACHE_KEY_1
          value: b&MTCiO=[T66RXm!jO@
        - name: CACHE_KEY_2
          value: PCAILGej5Ld@Q%{Q1=#
        - name: CACHE_KEY_3
          value: 2qz-]2OJlWDSTn_;RFQ
        image: nginx:1.17.3-alpine
        name: holy-api-container
        securityContext:                    # add
          allowPrivilegeEscalation: false   # add
          privileged: false                 # add
        volumeMounts:
        - mountPath: /cache1
          name: cache-volume1
        - mountPath: /cache2
          name: cache-volume2
        - mountPath: /cache3
          name: cache-volume3
      volumes:
      - emptyDir: {}
        name: cache-volume1
      - emptyDir: {}
        name: cache-volume2
      - emptyDir: {}
        name: cache-volume3
```

To indent multiple lines using `vim` you should set the shiftwidth using `:set shiftwidth=2`. Then mark multiple lines using `Shift v` and the up/down keys.

To then indent the marked lines press `>` or `<` and to repeat the action press `.`

Next create the new *Deployment*:

```
k -f /opt/course/9/holy-api-deployment.yaml create
```

and confirm it's running:

```
➜ k -n pluto get pod | grep holy
NAME                         READY   STATUS    RESTARTS   AGE
holy-api                     1/1     Running   0          19m
holy-api-5dbfdb4569-8qr5x    1/1     Running   0          30s
holy-api-5dbfdb4569-b5clh    1/1     Running   0          30s
holy-api-5dbfdb4569-rj2gz    1/1     Running   0          30s
```

Finally delete the single *Pod*:

```
k -n pluto delete pod holy-api --force --grace-period=0
```

```
➜ k -n pluto get pod,deployment | grep holy
pod/holy-api-5dbfdb4569-8qr5x   1/1     Running   0          2m4s
pod/holy-api-5dbfdb4569-b5clh   1/1     Running   0          2m4s
pod/holy-api-5dbfdb4569-rj2gz   1/1     Running   0          2m4s

deployment.extensions/holy-api   3/3     3            3           2m4s
```

Question 10 | Service, Logs

Solve this question on instance: `ssh ckad9043`

Team Pluto needs a new cluster internal *Service*. Create a ClusterIP *Service* named `project-plt-6cc-svc` in *Namespace* `pluto`. This *Service* should expose a single *Pod* named `project-plt-6cc-api` of image `nginx:1.17.3-alpine`, create that *Pod* as well. The *Pod* should be identified by label `project: plt-6cc-api`. The *Service* should use tcp port redirection of `3333:80`.

Finally use for example `curl` from a temporary `nginx:alpine` *Pod* to get the response from the *Service*. Write the response into `/opt/course/10/service_test.html` on `ckad9043`. Also check if the logs of *Pod* `project-plt-6cc-api` show the request and write those into `/opt/course/10/service_test.log` on `ckad9043`.

Answer

```
k -n pluto run project-plt-6cc-api --image=nginx:1.17.3-alpine --labels project=plt-6cc-api
```

This will create the requested *Pod*. In yaml it would look like this:

```
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    project: plt-6cc-api
  name: project-plt-6cc-api
spec:
  containers:
  - image: nginx:1.17.3-alpine
    name: project-plt-6cc-api
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
```

Next we create the service:

```
k -n pluto expose pod -h # help

k -n pluto expose pod project-plt-6cc-api --name project-plt-6cc-svc --port 3333 --target-port 80
```

Expose will create a yaml where everything is already set for our case and no need to change anything:

```
apiVersion: v1
kind: Service
metadata:
  creationTimestamp: null
  labels:
    project: plt-6cc-api
  name: project-plt-6cc-svc   # good
  namespace: pluto            # great
spec:
  ports:
  - port: 3333                # awesome
    protocol: TCP
    targetPort: 80            # nice
  selector:
```

```
    project: plt-6cc-api      # beautiful
status:
  loadBalancer: {}
```

We could also use `create service` but then we would need to change the yaml afterwards:

```
k -n pluto create service -h # help
k -n pluto create service clusterip -h #help
k -n pluto create service clusterip project-plt-6cc-svc --tcp 3333:80 --dry-run=client -oyaml
# now we would need to set the correct selector labels
```

Check the *Service* is running:

```
➜ k -n pluto get pod,svc | grep 6cc
pod/project-plt-6cc-api         1/1     Running   0          9m42s

service/project-plt-6cc-svc   ClusterIP   10.31.241.234   <none>        3333/TCP   2m24s
```

Does the *Service* has one *Endpoint*?

```
➜ k -n pluto describe svc project-plt-6cc-svc
Name:              project-plt-6cc-svc
Namespace:         pluto
Labels:            project=plt-6cc-api
Annotations:       <none>
Selector:          project=plt-6cc-api
Type:              ClusterIP
IP:                10.3.244.240
Port:              <unset>  3333/TCP
TargetPort:        80/TCP
Endpoints:         10.28.2.32:80
Session Affinity:  None
Events:            <none>
```

Or even shorter:

```
➜ k -n pluto get ep
NAME                  ENDPOINTS       AGE
project-plt-6cc-svc   10.28.2.32:80   84m
```

Yes, endpoint there! Finally we check the connection using a temporary *Pod*:

```
➜ k run tmp --restart=Never --rm --image=nginx:alpine -i -- curl http://project-plt-6cc-svc.pluto:3333
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   612  100   612    0     0  32210      0 --:--:-- --:--:-- --:--:-- 32210
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
    body {
        width: 35em;
        margin: 0 auto;
        font-family: Tahoma, Verdana, Arial, sans-serif;
    }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
...
```

Great! Notice that we use the Kubernetes *Namespace* dns resolving (`project-plt-6cc-svc.pluto`) here. We could only use the *Service* name if we would also spin up the temporary *Pod* in *Namespace* `pluto` .

And now really finally copy or pipe the html content into `/opt/course/10/service_test.html`.

```
# /opt/course/10/service_test.html
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
    body {
        width: 35em;
        margin: 0 auto;
        font-family: Tahoma, Verdana, Arial, sans-serif;
    }
...
```

Also the requested logs:

```
k -n pluto logs project-plt-6cc-api > /opt/course/10/service_test.log
```

```
# /opt/course/10/service_test.log
10.44.0.0 - - [22/Jan/2021:23:19:55 +0000] "GET / HTTP/1.1" 200 612 "-" "curl/7.69.1" "-"
```

Question 11 | Working with Containers

Solve this question on instance: `ssh ckad9043`

There are files to build a container image located at `/opt/course/11/image` on `ckad9043`. The container will run a Golang application which outputs information to stdout. You're asked to perform the following tasks:

> ℹ️ Run all Docker and Podman commands as user root. Use `sudo docker` and `sudo podman` or become root with `sudo -i`

1. Change the Dockerfile: set ENV variable `SUN_CIPHER_ID` to hardcoded value `5b9c1065-e39d-4a43-a04a-e59bcea3e03f`
2. Build the image using `sudo docker`, tag it `registry.killer.sh:5000/sun-cipher:v1-docker` and push it to the registry
3. Build the image using `sudo podman`, tag it `registry.killer.sh:5000/sun-cipher:v1-podman` and push it to the registry
4. Run a container using `sudo podman`, which keeps running detached in the background, named `sun-cipher` using image `registry.killer.sh:5000/sun-cipher:v1-podman`
5. Write the logs your container `sun-cipher` produces into `/opt/course/11/logs` on `ckad9043`

Answer

Dockerfile: list of commands from which an *Image* can be build

Image: binary file which includes all data/requirements to be run as a *Container*

Container: running instance of an *Image*

Registry: place where we can push/pull *Images* to/from

Step 1

We should probably create a backup:

```
➜ cp /opt/course/11/image/Dockerfile /opt/course/11/image/Dockerfile_bak
```

First we need to change the `/opt/course/11/logs/Dockerfile` to:

```
# build container stage 1
FROM docker.io/library/golang:1.15.15-alpine3.14
WORKDIR /src
COPY . .
RUN CGO_ENABLED=0 GOOS=linux go build -a -installsuffix cgo -o bin/app .

# app container stage 2
FROM docker.io/library/alpine:3.12.4
COPY --from=0 /src/bin/app app
# CHANGE NEXT LINE
ENV SUN_CIPHER_ID=5b9c1065-e39d-4a43-a04a-e59bcea3e03f
CMD ["./app"]
```

Step 2

Then we build the image using Docker:

```
➜ cd /opt/course/11/image

➜ sudo docker build -t registry.killer.sh:5000/sun-cipher:v1-docker .
...
Successfully built 409fde3c5bf9
Successfully tagged registry.killer.sh:5000/sun-cipher:v1-docker

➜ sudo docker image ls
REPOSITORY                            TAG          IMAGE ID       CREATED           SIZE
registry.killer.sh:5000/sun-cipher    v1-docker    409fde3c5bf9   24 seconds ago    7.76MB
...

➜ sudo docker push registry.killer.sh:5000/sun-cipher:v1-docker
The push refers to repository [registry.killer.sh:5000/sun-cipher]
c947fb5eba52: Pushed
```

```
33e8713114f8: Pushed
latest: digest: sha256:d216b4136a5b232b738698e826e7d12fccba9921d163b63777be23572250f23d size: 739
```

There we go, built and pushed.

Step 3

Next we build the image using Podman. Here it's only required to create one tag. The usage of Podman is very similar (for most cases even identical) to Docker:

```
➜ cd /opt/course/11/image

➜ sudo podman build -t registry.killer.sh:5000/sun-cipher:v1-podman .
...
--> 38adc53bd92
Successfully tagged registry.killer.sh:5000/sun-cipher:v1-podman
38adc53bd92881d91981c4b537f4f1b64f8de1de1b32eacc8479883170cee537

➜ sudo podman image ls
REPOSITORY                           TAG         IMAGE ID      CREATED         SIZE
registry.killer.sh:5000/sun-cipher   v1-podman   38adc53bd928  2 minutes ago  8.03 MB
...

➜ sudo podman push registry.killer.sh:5000/sun-cipher:v1-podman
Getting image source signatures
Copying blob 4d0d60db9eb6 done
Copying blob 33e8713114f8 done
Copying config bfa1a225f8 done
Writing manifest to image destination
Storing signatures
```

Built and pushed using Podman.

Step 4

We'll create a container from the perviously created image, using Podman, which keeps running in the background:

```
➜ sudo podman run -d --name sun-cipher registry.killer.sh:5000/sun-cipher:v1-podman
f8199cba792f9fd2d1bd4decc9b7a9c0acfb975d95eda35f5f583c9efbf95589
```

Step 5

Finally we need to collect some information into files:

```
➜ sudo podman logs sun-cipher
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 8081
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 7887
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 1847
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 4059
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 2081
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 1318
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 4425
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 2540
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 456
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 3300
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 694
2077/03/13 06:50:34 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 8511
2077/03/13 06:50:44 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 8162
2077/03/13 06:50:54 random number for 5b9c1065-e39d-4a43-a04a-e59bcea3e03f is 5089

➜ sudo podman logs sun-cipher > /opt/course/11/logs
```

This is looking not too bad at all. Our container skills are back in town!

Question 12 | Storage, PV, PVC, Pod volume

Solve this question on instance: `ssh ckad5601`

Create a new *PersistentVolume* named `earth-project-earthflower-pv`. It should have a capacity of *2Gi*, accessMode *ReadWriteOnce*, hostPath `/Volumes/Data` and no storageClassName defined.

Next create a new *PersistentVolumeClaim* in *Namespace* `earth` named `earth-project-earthflower-pvc` . It should request *2Gi* storage, accessMode *ReadWriteOnce* and should not define a storageClassName. The *PVC* should bound to the *PV* correctly.

Finally create a new *Deployment* `project-earthflower` in *Namespace* `earth` which mounts that volume at `/tmp/project-data`. The *Pods* of that *Deployment* should be of image `httpd:2.4.41-alpine`.

Answer

```
vim 12_pv.yaml
```

Find an example from https://kubernetes.io/docs and alter it:

```
# 12_pv.yaml
kind: PersistentVolume
apiVersion: v1
metadata:
 name: earth-project-earthflower-pv
spec:
 capacity:
  storage: 2Gi
 accessModes:
  - ReadWriteOnce
 hostPath:
  path: "/Volumes/Data"
```

Then create it:

```
k -f 12_pv.yaml create
```

Next the *PersistentVolumeClaim*:

```
vim 12_pvc.yaml
```

Find an example from https://kubernetes.io/docs and alter it:

```yaml
# 12_pvc.yaml
kind: PersistentVolumeClaim
apiVersion: v1
metadata:
  name: earth-project-earthflower-pvc
  namespace: earth
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
     storage: 2Gi
```

Then create:

```
k -f 12_pvc.yaml create
```

And check that both have the status Bound:

```
➜ k -n earth get pv,pvc
NAME                                  CAPACITY   ACCESS MODES   ...   STATUS   CLAIM
persistentvolume/...earthflower-pv    2Gi        RWO            ...   Bound    ...er-pvc

NAME                                          STATUS   VOLUME                          CAPACITY
persistentvolumeclaim/...earthflower-pvc      Bound    earth-project-earthflower-pv    2Gi
```

Next we create a *Deployment* and mount that volume:

```
k -n earth create deploy project-earthflower --image=httpd:2.4.41-alpine --dry-run=client -oyaml > 12_dep.yaml

vim 12_dep.yaml
```

Alter the yaml to mount the volume:

```yaml
# 12_dep.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  labels:
    app: project-earthflower
  name: project-earthflower
```

```
  namespace: earth
spec:
  replicas: 1
  selector:
    matchLabels:
      app: project-earthflower
  strategy: {}
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: project-earthflower
    spec:
      volumes:                                    # add
      - name: data                                # add
        persistentVolumeClaim:                    # add
          claimName: earth-project-earthflower-pvc  # add
      containers:
      - image: httpd:2.4.41-alpine
        name: container
        volumeMounts:                             # add
        - name: data                              # add
          mountPath: /tmp/project-data            # add
```

```
k -f 12_dep.yaml create
```

We can confirm it's mounting correctly:

```
➜ k -n earth describe pod project-earthflower-d6887f7c5-pn5wv | grep -A2 Mounts:
    Mounts:
      /tmp/project-data from data (rw) # there it is
      /var/run/secrets/kubernetes.io/serviceaccount from default-token-n2sjj (ro)
```

Question 13 | Storage, StorageClass, PVC

Solve this question on instance: `ssh ckad9043`

Team Moonpie, which has the *Namespace* `moon`, needs more storage. Create a new *PersistentVolumeClaim* named `moon-pvc-126` in that namespace. This claim should use a new *StorageClass* `moon-retain` with the *provisioner* set to `moon-retainer` and the *reclaimPolicy* set to *Retain*. The claim should request storage of *3Gi*, an *accessMode* of *ReadWriteOnce* and should use the new *StorageClass*.

The provisioner `moon-retainer` will be created by another team, so it's expected that the *PVC* will not boot yet. Confirm this by writing the event message from the *PVC* into file `/opt/course/13/pvc-126-reason` on `ckad9043`.

Answer

```
vim 13_sc.yaml
```

Head to https://kubernetes.io/docs, search for "storageclass" and alter the example code to this:

```
# 13_sc.yaml
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: moon-retain
provisioner: moon-retainer
reclaimPolicy: Retain
```

```
k create -f 13_sc.yaml
```

Now the same for the *PersistentVolumeClaim*, head to the docs, copy an example and transform it into:

```
vim 13_pvc.yaml
```

```
# 13_pvc.yaml
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: moon-pvc-126            # name as requested
  namespace: moon               # important
spec:
  accessModes:
    - ReadWriteOnce             # RWO
  resources:
    requests:
      storage: 3Gi              # size
  storageClassName: moon-retain # uses our new storage class
```

```
k -f 13_pvc.yaml create
```

Next we check the status of the *PVC* :

```
➜ k -n moon get pvc
NAME           STATUS    VOLUME   CAPACITY   ACCESS MODES   STORAGECLASS   AGE
moon-pvc-126   Pending                                      moon-retain    2m57s
```

```
➜ k -n moon describe pvc moon-pvc-126
Name:          moon-pvc-126
...
Status:        Pending
...
Events:
  Type    Reason                Age                  From                         Message
  ----    ------                ----                 ----                         -------
Normal  ExternalProvisioning  4s (x19 over 4m28s)  persistentvolume-controller    Waiting for a volume to be created
either by the external provisioner 'moon-retainer' or manually by the system administrator. If volume creation is
delayed, please verify that the provisioner is running and correctly registered.
```

This confirms that the *PVC* waits for the provisioner `moon-retainer` to be created. Finally we copy or write the event message into the requested location:

```
# /opt/course/13/pvc-126-reason
Waiting for a volume to be created either by the external provisioner 'moon-retainer' or manually by the system
administrator. If volume creation is delayed, please verify that the provisioner is running and correctly registered.
```

Question 14 | Secret, Secret-Volume, Secret-Env

Solve this question on instance: `ssh ckad9043`

You need to make changes on an existing *Pod* in *Namespace* `moon` called `secret-handler`. Create a new *Secret* `secret1` which contains `user=test` and `pass=pwd`. The *Secret*'s content should be available in *Pod* `secret-handler` as environment variables `SECRET1_USER` and `SECRET1_PASS`. The yaml for *Pod* `secret-handler` is available at `/opt/course/14/secret-handler.yaml`.

There is existing yaml for another *Secret* at `/opt/course/14/secret2.yaml`, create this *Secret* and mount it inside the same *Pod* at `/tmp/secret2`. Your changes should be saved under `/opt/course/14/secret-handler-new.yaml` on `ckad9043`. Both *Secrets* should only be available in *Namespace* `moon`.

Answer

```
k -n moon get pod # show pods
k -n moon create secret -h # help
k -n moon create secret generic -h # help
k -n moon create secret generic secret1 --from-literal user=test --from-literal pass=pwd
```

The last command would generate this yaml:

```
apiVersion: v1
data:
  pass: cHdk
  user: dGVzdA==
kind: Secret
metadata:
  creationTimestamp: null
  name: secret1
  namespace: moon
```

Next we create the second *Secret* from the given location, making sure it'll be created in *Namespace* `moon`:

```
k -n moon -f /opt/course/14/secret2.yaml create
```

```
➜ k -n moon get secret
NAME                  TYPE                                  DATA   AGE
default-token-rvzcf   kubernetes.io/service-account-token   3      66m
secret1               Opaque                                2      4m3s
secret2               Opaque                                1      8s
```

We will now edit the *Pod* yaml:

```
cp /opt/course/14/secret-handler.yaml /opt/course/14/secret-handler-new.yaml
vim /opt/course/14/secret-handler-new.yaml
```

Add the following to the yaml:

```yaml
# /opt/course/14/secret-handler-new.yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    id: secret-handler
    uuid: 1428721e-8d1c-4c09-b5d6-afd79200c56a
    red_ident: 9cf7a7c0-fdb2-4c35-9c13-c2a0bb52b4a9
    type: automatic
  name: secret-handler
  namespace: moon
spec:
  volumes:
  - name: cache-volume1
    emptyDir: {}
  - name: cache-volume2
    emptyDir: {}
  - name: cache-volume3
    emptyDir: {}
  - name: secret2-volume              # add
    secret:                           # add
      secretName: secret2             # add
  containers:
  - name: secret-handler
    image: bash:5.0.11
    args: ['bash', '-c', 'sleep 2d']
    volumeMounts:
    - mountPath: /cache1
      name: cache-volume1
    - mountPath: /cache2
      name: cache-volume2
    - mountPath: /cache3
      name: cache-volume3
    - name: secret2-volume            # add
      mountPath: /tmp/secret2         # add
    env:
    - name: SECRET_KEY_1
      value: ">8$kH#kj..i8}HImQd{"
    - name: SECRET_KEY_2
      value: "IO=a4L/XkRdvN8jM=Y+"
    - name: SECRET_KEY_3
```

```
      value: "-7PAO_Z]>{pwa43r)__"
    - name: SECRET1_USER              # add
      valueFrom:                      # add
        secretKeyRef:                 # add
          name: secret1               # add
          key: user                   # add
    - name: SECRET1_PASS              # add
      valueFrom:                      # add
        secretKeyRef:                 # add
          name: secret1               # add
          key: pass                   # add
```

There is also the possibility to import all keys from a *Secret* as env variables at once, though the env variable names will then be the same as in the *Secret*, which doesn't work for the requirements here:

```
  containers:
  - name: secret-handler
...
    envFrom:
    - secretRef:        # also works for configMapRef
        name: secret1
```

Then we apply the changes:

```
k -f /opt/course/14/secret-handler.yaml delete --force --grace-period=0
k -f /opt/course/14/secret-handler-new.yaml create
```

Instead of running delete and create we can also use recreate:

```
k -f /opt/course/14/secret-handler-new.yaml replace --force --grace-period=0
```

It was not requested directly, but you should always confirm it's working:

```
➜ k -n moon exec secret-handler -- env | grep SECRET1
SECRET1_USER=test
SECRET1_PASS=pwd

➜ k -n moon exec secret-handler -- find /tmp/secret2
/tmp/secret2
/tmp/secret2/..data
/tmp/secret2/key
/tmp/secret2/..2019_09_11_09_03_08.147048594
/tmp/secret2/..2019_09_11_09_03_08.147048594/key

➜ k -n moon exec secret-handler -- cat /tmp/secret2/key
12345678
```

Question 15 | ConfigMap, Configmap-Volume

Solve this question on instance: `ssh ckad9043`

Team Moonpie has a nginx server *Deployment* called `web-moon` in *Namespace* `moon`. Someone started configuring it but it was never completed. To complete please create a *ConfigMap* called `configmap-web-moon-html` containing the content of file `/opt/course/15/web-moon.html` under the data key-name `index.html`.

The *Deployment* `web-moon` is already configured to work with this *ConfigMap* and serve its content. Test the nginx configuration for example using `curl` from a temporary `nginx:alpine` *Pod*.

Answer

Let's check the existing *Pods*:

```
➜ k -n moon get pod
NAME                          READY   STATUS              RESTARTS   AGE
secret-handler                1/1     Running             0          55m
web-moon-847496c686-2rzj4     0/1     ContainerCreating   0          33s
web-moon-847496c686-9nwwj     0/1     ContainerCreating   0          33s
web-moon-847496c686-cxdbx     0/1     ContainerCreating   0          33s
web-moon-847496c686-hvqlw     0/1     ContainerCreating   0          33s
web-moon-847496c686-tj7ct     0/1     ContainerCreating   0          33s
```

```
➜ k -n moon describe pod web-moon-847496c686-2rzj4
...
Warning  FailedMount  31s (x7 over 63s)  kubelet, gke-test-default-pool-ce83a51a-p6s4  MountVolume.SetUp failed for volume "html-volume" : configmaps "configmap-web-moon-html" not found
```

Good so far, now let's create the missing *ConfigMap*:

```
k -n moon create configmap -h # help

k -n moon create configmap configmap-web-moon-html --from-file=index.html=/opt/course/15/web-moon.html # important to set the index.html key
```

This should create a *ConfigMap* with yaml like:

```
apiVersion: v1
data:
  index.html: |     # notice the key index.html, this will be the filename when mounted
    <!DOCTYPE html>
    <html lang="en">
    <head>
        <meta charset="UTF-8">
        <title>Web Moon Webpage</title>
    </head>
    <body>
    This is some great content.
    </body>
    </html>
kind: ConfigMap
metadata:
  creationTimestamp: null
  name: configmap-web-moon-html
  namespace: moon
```

After waiting a bit or deleting/recreating (`k -n moon rollout restart deploy web-moon`) the *Pods* we should see:

```
➜ k -n moon get pod
NAME                          READY   STATUS    RESTARTS   AGE
secret-handler                1/1     Running   0          59m
web-moon-847496c686-2rzj4     1/1     Running   0          4m28s
web-moon-847496c686-9nwwj     1/1     Running   0          4m28s
web-moon-847496c686-cxdbx     1/1     Running   0          4m28s
web-moon-847496c686-hvqlw     1/1     Running   0          4m28s
web-moon-847496c686-tj7ct     1/1     Running   0          4m28s
```

Looking much better. Finally we check if the nginx returns the correct content:

```
k -n moon get pod -o wide # get pod cluster IPs
```

Then use one IP to test the configuration:

```
➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl 10.44.0.78
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   161  100   161    0     0  80500      0 --:--:-- --:--:-- --:--:--  157k
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Web Moon Webpage</title>
</head>
<body>
This is some great content.
</body>
```

For debugging or further checks we could find out more about the *Pods* volume mounts:

```
➜ k -n moon describe pod web-moon-c77655cc-dc8v4 | grep -A2 Mounts:
    Mounts:
      /usr/share/nginx/html from html-volume (rw)
      /var/run/secrets/kubernetes.io/serviceaccount from default-token-rvzcf (ro)
```

And check the mounted folder content:

```
➜ k -n moon exec web-moon-c77655cc-dc8v4 find /usr/share/nginx/html
/usr/share/nginx/html
/usr/share/nginx/html/..2019_09_11_10_05_56.336284411
/usr/share/nginx/html/..2019_09_11_10_05_56.336284411/index.html
/usr/share/nginx/html/..data
/usr/share/nginx/html/index.html
```

Here it was important that the file will have the name `index.html` and not the original one `web-moon.html` which is controlled through the *ConfigMap* data key.

Question 16 | Logging sidecar

Solve this question on instance: `ssh ckad7326`

The Tech Lead of Mercury2D decided it's time for more logging, to finally fight all these missing data incidents. There is an existing container named `cleaner-con` in *Deployment* `cleaner` in *Namespace* `mercury`. This container mounts a volume and writes logs into a file called `cleaner.log`.

The yaml for the existing *Deployment* is available at `/opt/course/16/cleaner.yaml`. Persist your changes at `/opt/course/16/cleaner-new.yaml` on `ckad7326` but also make sure the *Deployment* is running.

Create a sidecar container named `logger-con`, image `busybox:1.31.0` , which mounts the same volume and writes the content of `cleaner.log` to stdout, you can use the `tail -f` command for this. This way it can be picked up by `kubectl logs`.

Check if the logs of the new container reveal something about the missing data incidents.

Answer

Sidecar containers in K8s are `initContainers` with `restartPolicy: Always`. Search for "Sidecar Containers" in the K8s Docs to familiarise yourself if necessary.

```
cp /opt/course/16/cleaner.yaml /opt/course/16/cleaner-new.yaml
vim /opt/course/16/cleaner-new.yaml
```

Add a sidecar container which outputs the log file to stdout:

```
# /opt/course/16/cleaner-new.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  name: cleaner
  namespace: mercury
spec:
  replicas: 2
  selector:
    matchLabels:
      id: cleaner
  template:
    metadata:
      labels:
        id: cleaner
    spec:
      volumes:
      - name: logs
        emptyDir: {}
      initContainers:
      - name: init
        image: bash:5.0.11
        command: ['bash', '-c', 'echo init > /var/log/cleaner/cleaner.log']
        volumeMounts:
        - name: logs
          mountPath: /var/log/cleaner
      - name: logger-con                                                  # add
        image: busybox:1.31.0                                             # add
        restartPolicy: Always                                             # add
        command: ["sh", "-c", "tail -f /var/log/cleaner/cleaner.log"]     # add
        volumeMounts:                                                     # add
        - name: logs                                                      # add
          mountPath: /var/log/cleaner                                     # add
      containers:
      - name: cleaner-con
        image: bash:5.0.11
        args: ['bash', '-c', 'while true; do echo `date`: "remove random file" >> /var/log/cleaner/cleaner.log; sleep
1; done']
        volumeMounts:
        - name: logs
          mountPath: /var/log/cleaner
```

In earlier K8s versions it was necessary to define sidecar containers as additional application containers under `containers:` like this:

```
# LEGACY example of defining sidecar containers in earlier K8s versions
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  name: cleaner
  namespace: mercury
spec:
...
  template:
...
    spec:
...
      initContainers:
      - name: init
        image: bash:5.0.11
...
      containers:
      - name: cleaner-con
        image: bash:5.0.11
...
      - name: logger-con                                                  # LEGACY example
        image: busybox:1.31.0                                             # LEGACY example
        command: ["sh", "-c", "tail -f /var/log/cleaner/cleaner.log"]     # LEGACY example
        volumeMounts:                                                     # LEGACY example
        - name: logs                                                      # LEGACY example
          mountPath: /var/log/cleaner                                     # LEGACY example
```

Then apply the changes and check the logs of the sidecar:

```
k -f /opt/course/16/cleaner-new.yaml apply
```

This will cause a deployment rollout of which we can get more details:

```
k -n mercury rollout history deploy cleaner
k -n mercury rollout history deploy cleaner --revision 1
k -n mercury rollout history deploy cleaner --revision 2
```

Check *Pod* statuses:

```
➜ k -n mercury get pod
NAME                       READY   STATUS        RESTARTS   AGE
cleaner-86b7758668-9pw6t   2/2     Running       0          6s
cleaner-86b7758668-qgh4v   0/2     Init:0/1      0          1s

➜ k -n mercury get pod
NAME                       READY   STATUS        RESTARTS   AGE
cleaner-86b7758668-9pw6t   2/2     Running       0          14s
cleaner-86b7758668-qgh4v   2/2     Running       0          9s
```

Finally check the logs of the logging sidecar container:

```
➜ k -n mercury logs cleaner-576967576c-cqtgx -c logger-con
init
Wed Sep 11 10:45:44 UTC 2099: remove random file
Wed Sep 11 10:45:45 UTC 2099: remove random file
...
```

Mystery solved, something is removing files at random ;) It's important to understand how containers can communicate with each other using volumes.

Question 17 | InitContainer

Solve this question on instance: `ssh ckad5601`

Last lunch you told your coworker from department Mars Inc how amazing *InitContainer*s are. Now he would like to see one in action. There is a *Deployment* yaml at `/opt/course/17/test-init-container.yaml`. This *Deployment* spins up a single *Pod* of image `nginx:1.17.3-alpine` and serves files from a mounted volume, which is empty right now.

Create an *InitContainer* named `init-con` which also mounts that volume and creates a file `index.html` with content `check this out!` in the root of the mounted volume. For this test we ignore that it doesn't contain valid html.

The *InitContainer* should be using image `busybox:1.31.0`. Test your implementation for example using `curl` from a temporary `nginx:alpine` *Pod*.

Answer

```
cp /opt/course/17/test-init-container.yaml ~/17_test-init-container.yaml

vim 17_test-init-container.yaml
```

Add the *InitContainer*:

```
# 17_test-init-container.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: test-init-container
  namespace: mars
spec:
  replicas: 1
  selector:
    matchLabels:
      id: test-init-container
  template:
    metadata:
      labels:
        id: test-init-container
    spec:
      volumes:
      - name: web-content
        emptyDir: {}
      initContainers:                # initContainer start
      - name: init-con
        image: busybox:1.31.0
        command: ['sh', '-c', 'echo "check this out!" > /tmp/web-content/index.html']
        volumeMounts:
        - name: web-content
          mountPath: /tmp/web-content # initContainer end
      containers:
      - image: nginx:1.17.3-alpine
        name: nginx
        volumeMounts:
        - name: web-content
          mountPath: /usr/share/nginx/html
        ports:
        - containerPort: 80
```

Then we create the *Deployment*:

```
k -f 17_test-init-container.yaml create
```

Finally we test the configuration:

```
k -n mars get pod -o wide # to get the cluster IP
```

```
➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl 10.0.0.67
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
check this out!
```

Beautiful.

Question 18 | Service misconfiguration

Solve this question on instance: `ssh ckad5601`

There seems to be an issue in *Namespace* `mars` where the ClusterIP service `manager-api-svc` should make the *Pods* of *Deployment* `manager-api-deployment` available inside the cluster.

You can test this with `curl manager-api-svc.mars:4444` from a temporary `nginx:alpine` *Pod*. Check for the misconfiguration and apply a fix.

Answer

First let's get an overview:

```
➜ k -n mars get all
NAME                                        READY   STATUS    RESTARTS   AGE
pod/manager-api-deployment-dbcc6657d-bg2hh   1/1     Running   0          98m
pod/manager-api-deployment-dbcc6657d-f5fv4   1/1     Running   0          98m
pod/manager-api-deployment-dbcc6657d-httjv   1/1     Running   0          98m
pod/manager-api-deployment-dbcc6657d-k98xn   1/1     Running   0          98m
```

```
pod/test-init-container-5db7c99857-htx6b     1/1     Running   0          2m19s

NAME                         TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
service/manager-api-svc      ClusterIP   10.15.241.159   <none>        4444/TCP   99m

NAME                                       READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/manager-api-deployment     4/4     4            4           98m
deployment.apps/test-init-container        1/1     1            1           2m19s
...
```

Everything seems to be running, but we can't seem to get a connection:

```
➜ k -n mars run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 manager-api-svc:4444
If you don't see a command prompt, try pressing enter.
  0     0    0     0    0     0      0      0 --:--:--  0:00:01 --:--:--     0
curl: (28) Connection timed out after 1000 milliseconds
pod "tmp" deleted
pod mars/tmp terminated (Error)
```

Ok, let's try to connect to one pod directly:

```
k -n mars get pod -o wide # get cluster IP
```

```
➜ k -n mars run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 10.0.1.14
 % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
...
```

The *Pods* itself seem to work. Let's investigate the *Service* a bit:

```
➜ k -n mars describe service manager-api-svc
Name:              manager-api-svc
Namespace:         mars
Labels:            app=manager-api-svc
...
Endpoints:         <none>
...
```

Endpoint inspection is also possible using:

```
k -n mars get ep
```

No endpoints - No good. We check the *Service* yaml:

```
k -n mars edit service manager-api-svc
```

```
# k -n mars edit service manager-api-svc
apiVersion: v1
kind: Service
metadata:
...
  labels:
    app: manager-api-svc
  name: manager-api-svc
  namespace: mars
...
spec:
  clusterIP: 10.3.244.121
  ports:
  - name: 4444-80
    port: 4444
    protocol: TCP
    targetPort: 80
  selector:
    #id: manager-api-deployment # wrong selector, needs to point to pod!
    id: manager-api-pod
  sessionAffinity: None
  type: ClusterIP
```

Though *Pods* are usually never created without a *Deployment* or *ReplicaSet*, *Services* always select for *Pods* directly. This gives great flexibility because *Pods* could be created through various customized ways. After saving the new selector we check the *Service* again for endpoints:

```
➜ k -n mars get ep
NAME              ENDPOINTS                                         AGE
manager-api-svc   10.0.0.30:80,10.0.1.30:80,10.0.1.31:80 + 1 more...   41m
```

Endpoints - Good! Now we try connecting again:

```
➜ k -n mars run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 manager-api-svc:4444
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   612  100   612    0     0    99k      0 --:--:-- --:--:-- --:--:--   99k
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
...
```

And we fixed it. Good to know is how to be able to use Kubernetes DNS resolution from a different *Namespace*. Not necessary, but we could spin up the temporary *Pod* in default *Namespace*:

```
➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 manager-api-svc:4444
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0curl: (6) Could not resolve host:
manager-api-svc
pod "tmp" deleted
pod default/tmp terminated (Error)

➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 manager-api-svc.mars:4444
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   612  100   612    0     0  68000      0 --:--:-- --:--:-- --:--:-- 68000
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
```

Short `manager-api-svc.mars` or long `manager-api-svc.mars.svc.cluster.local` work.

Question 19 | Service ClusterIP->NodePort

Solve this question on instance: `ssh ckad5601`

In *Namespace* `jupiter` you'll find an apache *Deployment* (with one replica) named `jupiter-crew-deploy` and a ClusterIP *Service* called `jupiter-crew-svc` which exposes it. Change this service to a NodePort one to make it available on all nodes on port 30100.

Test the NodePort *Service* using the internal IP of all available nodes and the port 30100 using `curl`, you can reach the internal node IPs directly from your main terminal. On which nodes is the *Service* reachable? On which node is the *Pod* running?

Answer

First we get an overview:

```
➜ k -n jupiter get all
NAME                                     READY   STATUS    RESTARTS   AGE
pod/jupiter-crew-deploy-8cdf99bc9-klwqt   1/1     Running   0          34m

NAME                         TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
service/jupiter-crew-svc     ClusterIP   10.100.254.66   <none>        8080/TCP   34m
...
```

(Optional) Next we check if the ClusterIP *Service* actually works:

```
➜ k -n jupiter run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 jupiter-crew-svc:8080
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100    45  100    45    0     0   5000      0 --:--:-- --:--:-- --:--:--  5000
<html><body><h1>It works!</h1></body></html>
```

The *Service* is working great. Next we change the *Service* type to NodePort and set the port:

```
k -n jupiter edit service jupiter-crew-svc
```

```
# k -n jupiter edit service jupiter-crew-svc
apiVersion: v1
kind: Service
metadata:
  name: jupiter-crew-svc
  namespace: jupiter
...
spec:
  clusterIP: 10.3.245.70
  ports:
  - name: 8080-80
```

```
    port: 8080
    protocol: TCP
    targetPort: 80
    nodePort: 30100 # add the nodePort
  selector:
    id: jupiter-crew
  sessionAffinity: None
  #type: ClusterIP
  type: NodePort    # change type
status:
  loadBalancer: {}
```

We check if the *Service* type was updated:

```
➜ k -n jupiter get svc
NAME               TYPE       CLUSTER-IP    EXTERNAL-IP   PORT(S)          AGE
jupiter-crew-svc   NodePort   10.3.245.70   <none>        8080:30100/TCP   3m52s
```

(Optional) And we confirm that the service is still reachable internally:

```
➜ k -n jupiter run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 jupiter-crew-svc:8080
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
<html><body><h1>It works!</h1></body></html>
```

Nice. A NodePort *Service* kind of lies on top of a ClusterIP one, making the ClusterIP *Service* reachable on the Node IPs (internal and external). Next we get the *internal* IPs of all nodes to check the connectivity:

```
➜ k get nodes -o wide
NAME      STATUS   ROLES           AGE   VERSION   INTERNAL-IP      ...
ckad5601  Ready    control-plane   18h   v1.33.1   192.168.100.11   ...
```

We can test the connection using the node IP:

```
➜ curl 192.168.100.11:30100
<html><body><h1>It works!</h1></body></html>
```

Here we only have one node in the cluster, but the *Service* would be reachable on all of them. Even if the *Pod* is just running on one specific node, the *Service* makes it available through port 30100 on the internal and external IP addresses of all nodes. This is at least the common/default behaviour but can depend on cluster configuration.

Question 20 | NetworkPolicy

Solve this question on instance: `ssh ckad7326`

In *Namespace* `venus` you'll find two *Deployments* named `api` and `frontend`. Both *Deployments* are exposed inside the cluster using *Services*. Create a *NetworkPolicy* named `np1` which restricts outgoing tcp connections from *Deployment* `frontend` and only allows those going to *Deployment* `api`. Make sure the *NetworkPolicy* still allows outgoing traffic on UDP/TCP ports 53 for DNS resolution.

Test using: `wget www.google.com` and `wget api:2222` from a *Pod* of *Deployment* `frontend`.

Answer

> ***INFO:*** For learning NetworkPolicies check out https://editor.cilium.io. But you're not allowed to use it during the exam.

First we get an overview:

```
➜ k -n venus get all
NAME                            READY   STATUS    RESTARTS   AGE
pod/api-5979b95578-gktxp        1/1     Running   0          57s
pod/api-5979b95578-lhcl5        1/1     Running   0          57s
pod/frontend-789cbdc677-c9v8h   1/1     Running   0          57s
pod/frontend-789cbdc677-npk2m   1/1     Running   0          57s
pod/frontend-789cbdc677-pl67g   1/1     Running   0          57s
pod/frontend-789cbdc677-rjt5r   1/1     Running   0          57s
pod/frontend-789cbdc677-xgf5n   1/1     Running   0          57s

NAME               TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)    AGE
service/api        ClusterIP   10.3.255.137   <none>        2222/TCP   37s
service/frontend   ClusterIP   10.3.255.135   <none>        80/TCP     57s
...
```

(Optional) This is not necessary but we could check if the *Services* are working inside the cluster:

```
➜ k -n venus run tmp --restart=Never --rm -i --image=busybox -i -- wget -O- frontend:80
Connecting to frontend:80 (10.3.245.9:80)
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
...

➜ k -n venus run tmp --restart=Never --rm --image=busybox -i -- wget -O- api:2222
Connecting to api:2222 (10.3.250.233:2222)
<html><body><h1>It works!</h1></body></html>
```

Then we use any `frontend` *Pod* and check if it can reach external names and the `api` *Service*:

```
➜ k -n venus exec frontend-789cbdc677-c9v8h -- wget -O- www.google.com
Connecting to www.google.com (216.58.205.227:80)
-                    100% |********************************| 12955  0:00:00 ETA
<!doctype html><html itemscope="" itemtype="http://schema.org/WebPage" lang="en"><head>
...

➜ k -n venus exec frontend-789cbdc677-c9v8h -- wget -O- api:2222
<html><body><h1>It works!</h1></body></html>
Connecting to api:2222 (10.3.255.137:2222)
-                    100% |********************************|    45  0:00:00 ETA
...
```

We see *Pods* of `frontend` can reach the `api` and external names.

```
vim 20_np1.yaml
```

Now we head to https://kubernetes.io/docs, search for *NetworkPolicy*, copy the example code and adjust it to:

```
# 20_np1.yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: np1
  namespace: venus
spec:
  podSelector:
    matchLabels:
      id: frontend          # label of the pods this policy should be applied on
```

```
  policyTypes:
  - Egress                  # we only want to control egress
  egress:
  - to:                     # 1st egress rule
    - podSelector:            # allow egress only to pods with api label
        matchLabels:
          id: api
  - ports:                  # 2nd egress rule
    - port: 53                # allow DNS UDP
      protocol: UDP
    - port: 53                # allow DNS TCP
      protocol: TCP
```

Notice that we specify two egress rules in the yaml above. If we specify multiple egress rules then these are connected using a logical OR. So in the example above we do:

```
allow outgoing traffic if
  (destination pod has label id:api) OR ((port is 53 UDP) OR (port is 53 TCP))
```

Let's have a look at example code which wouldn't work in our case:

```
# this example does not work in our case
...
  egress:
  - to:                     # 1st AND ONLY egress rule
    - podSelector:            # allow egress only to pods with api label
        matchLabels:
          id: api
    ports:                  # STILL THE SAME RULE but just an additional selector
    - port: 53                # allow DNS UDP
      protocol: UDP
    - port: 53                # allow DNS TCP
      protocol: TCP
```

In the yaml above we only specify one egress rule with two selectors. It can be translated into:

```
allow outgoing traffic if
  (destination pod has label id:api) AND ((port is 53 UDP) OR (port is 53 TCP))
```

Apply the correct policy:

```
k -f 20_np1.yaml create
```

And try again, external is not working any longer:

```
➜ k -n venus exec frontend-789cbdc677-c9v8h -- wget -O- www.google.de
Connecting to www.google.de:2222 (216.58.207.67:80)
^C

➜ k -n venus exec frontend-789cbdc677-c9v8h -- wget -O- -T 5 www.google.de:80
Connecting to www.google.com (172.217.203.104:80)
wget: download timed out
command terminated with exit code 1
```

Internal connection to `api` work as before:

```
➜ k -n venus exec frontend-789cbdc677-c9v8h -- wget -O- api:2222
<html><body><h1>It works!</h1></body></html>
Connecting to api:2222 (10.3.255.137:2222)
-                    100% |********************************|    45  0:00:00 ETA
```

Question 21 | Requests and Limits, ServiceAccount

Solve this question on instance: `ssh ckad7326`

Team Neptune needs 3 *Pods* of image `httpd:2.4-alpine`, create a *Deployment* named `neptune-10ab` for this. The containers should be named `neptune-pod-10ab`. Each container should have a memory request of *20Mi* and a memory limit of *50Mi*.

Team Neptune has its own *ServiceAccount* `neptune-sa-v2` under which the *Pods* should run. The *Deployment* should be in *Namespace* `neptune`.

Answer:

```
k -n neptune create deployment -h # help
k -n neptune create deploy -h # deploy is short for deployment

k -n neptune create deploy neptune-10ab --replicas=3 --image=httpd:2.4-alpine --dry-run=client -oyaml > 21.yaml

vim 21.yaml
```

Now make the required changes using vim:

```
# 21.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  labels:
    app: neptune-10ab
  name: neptune-10ab
  namespace: neptune
spec:
  replicas: 3
  selector:
    matchLabels:
      app: neptune-10ab
  strategy: {}
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: neptune-10ab
    spec:
      serviceAccountName: neptune-sa-v2 # add
      containers:
      - image: httpd:2.4-alpine
        name: neptune-pod-10ab  # change
        resources:              # add
          limits:               # add
            memory: 50Mi        # add
          requests:             # add
            memory: 20Mi        # add
status: {}
```

Then create the yaml:

```
k create -f 21.yaml # namespace already set in yaml
```

To verify all *Pods* are running we do:

```
➜ k -n neptune get pod | grep neptune-10ab
neptune-10ab-7d4b8d45b-4nzj5   1/1     Running            0          57s
neptune-10ab-7d4b8d45b-lzwrf   1/1     Running            0          17s
neptune-10ab-7d4b8d45b-z5hcc   1/1     Running            0          17s
```

Question 22 | Labels, Annotations

Solve this question on instance: `ssh ckad9043`

Team Sunny needs to identify some of their *Pods* in namespace `sun`. They ask you to add a new label `protected: true` to all *Pods* with an existing label `type: worker` or `type: runner`. Also add an annotation `protected: do not delete this pod` to all *Pods* having the new label `protected: true`.

Answer

```
➜ k -n sun get pod --show-labels
NAME           READY   STATUS    RESTARTS   AGE   LABELS
0509649a       1/1     Running   0          25s   type=runner,type_old=messenger
0509649b       1/1     Running   0          24s   type=worker
1428721e       1/1     Running   0          23s   type=worker
1428721f       1/1     Running   0          22s   type=worker
43b9a          1/1     Running   0          22s   type=test
4c09           1/1     Running   0          21s   type=worker
4c35           1/1     Running   0          20s   type=worker
4fe4           1/1     Running   0          19s   type=worker
5555a          1/1     Running   0          19s   type=messenger
86cda          1/1     Running   0          18s   type=runner
8d1c           1/1     Running   0          17s   type=messenger
a004a          1/1     Running   0          16s   type=runner
```

```
a94128196      1/1     Running   0          15s   type=runner,type_old=messenger
afd79200c56a   1/1     Running   0          15s   type=worker
b667           1/1     Running   0          14s   type=worker
fdb2           1/1     Running   0          13s   type=worker
```

If we would only like to get pods with certain labels we can run:

```
k -n sun get pod -l type=runner # only pods with label runner
```

We can use this label filtering also when using other commands, like setting new labels:

```
k label -h # help
k -n sun label pod -l type=runner protected=true # run for label runner
k -n sun label pod -l type=worker protected=true # run for label worker
```

Or we could run:

```
k -n sun label pod -l "type in (worker,runner)" protected=true
```

Let's check the result:

```
→ k -n sun get pod --show-labels
NAME           ...   AGE   LABELS
0509649a       ...          56s   protected=true,type=runner,type_old=messenger
0509649b       ...          55s   protected=true,type=worker
1428721e       ...          54s   protected=true,type=worker
1428721f       ...          53s   protected=true,type=worker
43b9a          ...          53s   type=test
4c09           ...          52s   protected=true,type=worker
4c35           ...          51s   protected=true,type=worker
4fe4           ...          50s   protected=true,type=worker
5555a          ...          50s   type=messenger
86cda          ...          49s   protected=true,type=runner
8d1c           ...          48s   type=messenger
a004a          ...          47s   protected=true,type=runner
a94128196      ...          46s   protected=true,type=runner,type_old=messenger
afd79200c56a   ...          46s   protected=true,type=worker
b667           ...          45s   protected=true,type=worker
fdb2           ...          44s   protected=true,type=worker
```

Looking good. Finally we set the annotation using the newly assigned label `protected: true`:

```
k -n sun annotate pod -l protected=true protected="do not delete this pod"
```

Not requested in the task but for your own control you could run:

```
k -n sun get pod -l protected=true -o yaml | grep -A 8 metadata:
```

CKAD Simulator Preview Kubernetes 1.33

https://killer.sh

This is a preview of the CKAD Simulator content. The full CKAD Simulator contains 22 different questions. These preview questions are in addition to the provided ones and can also be solved in the interactive environment.

Preview Question 1

Solve this question on instance: `ssh ckad9043`

In *Namespace* `pluto` there is a *Deployment* named `project-23-api`. It has been working okay for a while but Team Pluto needs it to be more reliable. Implement a liveness-probe which checks the container to be reachable on port 80. Initially the probe should wait *10*, periodically *15* seconds.

The original *Deployment* yaml is available at `/opt/course/p1/project-23-api.yaml`. Save your changes at `/opt/course/p1/project-23-api-new.yaml` and apply the changes.

Answer

First we get an overview:

```
➜ k -n pluto get all -o wide
NAME                                 READY   STATUS    ... IP           ...
pod/holy-api                         1/1     Running   ... 10.12.0.26   ...
pod/project-23-api-784857f54c-dx6h6  1/1     Running   ... 10.12.2.15   ...
pod/project-23-api-784857f54c-sj8df  1/1     Running   ... 10.12.1.18   ...
pod/project-23-api-784857f54c-t4xmh  1/1     Running   ... 10.12.0.23   ...

NAME                                 READY   UP-TO-DATE   AVAILABLE   ...
deployment.apps/project-23-api       3/3     3            3           ...
```

To note: we see another *Pod* here called `holy-api` which is part of another section. This is often the case in the provided scenarios, so be careful to only manipulate the resources you need to. Just like in the real world and in the exam.

Next we use `nginx:alpine` and `curl` to check if one *Pod* is accessible on port 80:

```
➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 10.12.2.15
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
...
```

We could also use `busybox` and `wget` for this:

```
➜ k run tmp --restart=Never --rm --image=busybox -i -- wget -O- 10.12.2.15
Connecting to 10.12.2.15 (10.12.2.15:80)
writing to stdout
-                    100% |********************************|   612  0:00:00 ETA
written to stdout
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
```

Now that we're sure the *Deployment* works we can continue with altering the provided yaml:

```
cp /opt/course/p1/project-23-api.yaml /opt/course/p1/project-23-api-new.yaml
vim /opt/course/p1/project-23-api-new.yaml
```

Add the liveness-probe to the yaml:

```
# /opt/course/p1/project-23-api-new.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: project-23-api
  namespace: pluto
spec:
  replicas: 3
  selector:
    matchLabels:
      app: project-23-api
  template:
    metadata:
      labels:
        app: project-23-api
    spec:
      volumes:
      - name: cache-volume1
        emptyDir: {}
      - name: cache-volume2
        emptyDir: {}
      - name: cache-volume3
        emptyDir: {}
      containers:
      - image: httpd:2.4-alpine
        name: httpd
        volumeMounts:
        - mountPath: /cache1
          name: cache-volume1
        - mountPath: /cache2
          name: cache-volume2
        - mountPath: /cache3
          name: cache-volume3
        env:
        - name: APP_ENV
          value: "prod"
        - name: APP_SECRET_N1
          value: "IO=a4L/XkRdvN8jM=Y+"
        - name: APP_SECRET_P1
          value: "-7PAO_Z]>{pwa43r)__"
        livenessProbe:                  # add
          tcpSocket:                    # add
            port: 80                    # add
          initialDelaySeconds: 10       # add
          periodSeconds: 15             # add
```

Then let's apply the changes:

```
k -f /opt/course/p1/project-23-api-new.yaml apply
```

Next we wait 10 seconds and confirm the *Pods* are still running:

```
➜ k -n pluto get pod
NAME                              READY   STATUS    RESTARTS   AGE
holy-api                          1/1     Running   0          144m
project-23-api-5b4579fd49-8knh8   1/1     Running   0          90s
project-23-api-5b4579fd49-cbgph   1/1     Running   0          88s
project-23-api-5b4579fd49-tcfq5   1/1     Running   0          86s
```

We can also check the configured liveness-probe settings on a *Pod* or the *Deployment*:

```
➜ k -n pluto describe pod project-23-api-5b4579fd49-8knh8 | grep Liveness
    Liveness:   tcp-socket :80 delay=10s timeout=1s period=15s #success=1 #failure=3

➜  k -n pluto describe deploy project-23-api | grep Liveness
    Liveness:   tcp-socket :80 delay=10s timeout=1s period=15s #success=1 #failure=3
```

Preview Question 2

Solve this question on instance: `ssh ckad9043`

Team Sun needs a new *Deployment* named `sunny` with 4 replicas of image `nginx:1.17.3-alpine` in *Namespace* `sun`. The *Deployment* and its *Pods* should use the existing *ServiceAccount* `sa-sun-deploy`.

Expose the *Deployment* internally using a ClusterIP *Service* named `sun-srv` on port 9999. The nginx containers should run as default on port 80. The management of Team Sun would like to execute a command to check that all *Pods* are running on occasion. Write that command into file `/opt/course/p2/sunny_status_command.sh`. The command should use `kubectl`.

Answer

```
k -n sun create deployment -h #help

k -n sun create deployment sunny --image=nginx:1.17.3-alpine --dry-run=client -oyaml > p2_sunny.yaml

vim p2_sunny.yaml
```

Then alter its yaml to include the requirements:

```
# p2_sunny.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  labels:
    app: sunny
  name: sunny
  namespace: sun
spec:
  replicas: 4                                # change
  selector:
    matchLabels:
      app: sunny
  strategy: {}
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: sunny
    spec:
      serviceAccountName: sa-sun-deploy      # add
      containers:
      - image: nginx:1.17.3-alpine
        name: nginx
        resources: {}
status: {}
```

Now create the yaml and confirm it's running:

```
➜ k create -f p2_sunny.yaml
deployment.apps/sunny created

➜ k -n sun get pod
NAME                     READY   STATUS        RESTARTS   AGE
0509649a                 1/1     Running       0          149m
0509649b                 1/1     Running       0          149m
1428721e                 1/1     Running       0          149m
...
sunny-64df8dbdbb-9mxbw   1/1     Running       0          10s
sunny-64df8dbdbb-mp5cf   1/1     Running       0          10s
sunny-64df8dbdbb-pggdf   1/1     Running       0          6s
sunny-64df8dbdbb-zvqth   1/1     Running       0          7s
```

Confirmed, the AGE column is always in important information about if changes were applied. Next we expose the *Pods* by created the *Service*:

```
k -n sun expose -h # help
k -n sun expose deployment sunny --name sun-srv --port 9999 --target-port 80
```

Using expose instead of `kubectl create service clusterip` is faster because it already sets the correct selector-labels. The previous command would produce this yaml:

```
# k -n sun expose deployment sunny --name sun-srv --port 9999 --target-port 80
apiVersion: v1
kind: Service
metadata:
  creationTimestamp: null
  labels:
    app: sunny
  name: sun-srv         # required by task
spec:
  ports:
  - port: 9999          # service port
    protocol: TCP
    targetPort: 80      # target port
  selector:
    app: sunny          # selector is important
status:
  loadBalancer: {}
```

Let's test the *Service* using `wget` from a temporary *Pod*:

```
➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 sun-srv.sun:9999
Connecting to sun-srv.sun:9999 (10.23.253.120:9999)
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
...
```

Because the *Service* is in a different *Namespace* as our temporary *Pod*, it is reachable using the names `sun-srv.sun` or fully: `sun-srv.sun.svc.cluster.local`.

Finally we need a command which can be executed to check if all *Pods* are runing, this can be done with:

```
vim /opt/course/p2/sunny_status_command.sh
```

```
# /opt/course/p2/sunny_status_command.sh
kubectl -n sun get deployment sunny
```

To run the command:

```
➜ sh /opt/course/p2/sunny_status_command.sh
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
sunny   4/4     4            4           13m
```

Preview Question 3

Solve this question on instance: `ssh ckad5601`

Management of EarthAG recorded that one of their *Services* stopped working. Dirk, the administrator, left already for the long weekend. All the information they could give you is that it was located in *Namespace* `earth` and that it stopped working after the latest rollout. All *Services* of EarthAG should be reachable from inside the cluster.

Find the *Service*, fix any issues and confirm it's working again. Write the reason of the error into file `/opt/course/p3/ticket-654.txt` so Dirk knows what the issue was.

Answer

First we get an overview of the resources in *Namespace* `earth`:

```
➜ k -n earth get all
NAME                                          READY   STATUS    RESTARTS   AGE
pod/earth-2x3-api-584df69757-ngnwp            1/1     Running   0          116m
pod/earth-2x3-api-584df69757-ps8cs            1/1     Running   0          116m
pod/earth-2x3-api-584df69757-ww9q8            1/1     Running   0          116m
pod/earth-2x3-web-85c5b7986c-48vjt            1/1     Running   0          116m
pod/earth-2x3-web-85c5b7986c-6mqmb            1/1     Running   0          116m
pod/earth-2x3-web-85c5b7986c-6vjll            1/1     Running   0          116m
pod/earth-2x3-web-85c5b7986c-fnkbp            1/1     Running   0          116m
pod/earth-2x3-web-85c5b7986c-pjm5m            1/1     Running   0          116m
pod/earth-2x3-web-85c5b7986c-pwfvj            1/1     Running   0          116m
pod/earth-3cc-runner-6cb6cc6974-8wm5x         1/1     Running   0          116m
pod/earth-3cc-runner-6cb6cc6974-9fx8b         1/1     Running   0          116m
pod/earth-3cc-runner-6cb6cc6974-b9nrv         1/1     Running   0          116m
pod/earth-3cc-runner-heavy-6bf876f46d-b47vq   1/1     Running   0          116m
pod/earth-3cc-runner-heavy-6bf876f46d-mrzqd   1/1     Running   0          116m
pod/earth-3cc-runner-heavy-6bf876f46d-qkd74   1/1     Running   0          116m
pod/earth-3cc-web-6bfdf8b848-f74cj            0/1     Running   0          116m
pod/earth-3cc-web-6bfdf8b848-n4z7z            0/1     Running   0          116m
pod/earth-3cc-web-6bfdf8b848-rcmxs            0/1     Running   0          116m
pod/earth-3cc-web-6bfdf8b848-xl467            0/1     Running   0          116m


NAME                          TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)    AGE
service/earth-2x3-api-svc     ClusterIP   10.3.241.242   <none>        4546/TCP   116m
service/earth-2x3-web-svc     ClusterIP   10.3.250.247   <none>        4545/TCP   116m
service/earth-3cc-web         ClusterIP   10.3.243.24    <none>        6363/TCP   116m


NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/earth-2x3-api            3/3     3            3           116m
deployment.apps/earth-2x3-web            6/6     6            6           116m
deployment.apps/earth-3cc-runner         3/3     3            3           116m
deployment.apps/earth-3cc-runner-heavy   3/3     3            3           116m
deployment.apps/earth-3cc-web            0/4     4            0           116m

NAME                                                DESIRED   CURRENT   READY   AGE
```

```
replicaset.apps/earth-2x3-api-584df69757              3         3         3       116m
replicaset.apps/earth-2x3-web-85c5b7986c              6         6         6       116m
replicaset.apps/earth-3cc-runner-6cb6cc6974           3         3         3       116m
replicaset.apps/earth-3cc-runner-heavy-6bf876f46d     3         3         3       116m
replicaset.apps/earth-3cc-web-6895587dc7              0         0         0       116m
replicaset.apps/earth-3cc-web-6bfdf8b848              4         4         0       116m
replicaset.apps/earth-3cc-web-d49645966               0         0         0       116m
```

First impression could be that all *Pods* are in status RUNNING. But looking closely we see that some of the *Pods* are not ready, which also confirms what we see about one *Deployment* and one *ReplicaSet*. This could be our error to further investigate.

Another approach could be to check the *Service*s for missing endpoints:

```
➜ k -n earth get ep
NAME                ENDPOINTS                                      AGE
earth-2x3-api-svc   10.0.0.10:80,10.0.1.5:80,10.0.2.4:80           116m
earth-2x3-web-svc   10.0.0.11:80,10.0.0.12:80,10.0.1.6:80 + 3 more...   116m
earth-3cc-web
```

Service `earth-3cc-web` doesn't have endpoints. This could be a selector/label misconfiguration or the endpoints are actually not available/ready.

Checking all *Services* for connectivity should show the same (this step is optional and just for demonstration):

```
➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 earth-2x3-api-svc.earth:4546
...
<html><body><h1>It works!</h1></body></html>

➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 earth-2x3-web-svc.earth:4545
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100    45  100    45    0     0   5000      0 --:--:-- --:--:-- --:--:--  5000
<html><body><h1>It works!</h1></body></html>

➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 earth-3cc-web.earth:6363
If you don't see a command prompt, try pressing enter.
  0     0    0     0    0     0      0      0 --:--:--  0:00:05 --:--:--     0
curl: (28) Connection timed out after 5000 milliseconds
pod "tmp" deleted
pod default/tmp terminated (Error)
```

Notice that we use here for example `earth-2x3-api-svc.earth`. We could also spin up a temporary *Pod* in *Namespace* `earth` and connect directly to `earth-2x3-api-svc`.

We get no connection to `earth-3cc-web.earth:6363`. Let's look at the *Deployment* `earth-3cc-web`. Here we see that the requested amount of replicas is not available/ready:

```
➜ k -n earth get deploy earth-3cc-web
NAME            READY   UP-TO-DATE   AVAILABLE   AGE
earth-3cc-web   0/4     4            0           7m18s
```

To continue we check the *Deployment* yaml for some misconfiguration:

```
k -n earth edit deploy earth-3cc-web
```

```
# k -n earth edit deploy earth-3cc-web
apiVersion: extensions/v1beta1
kind: Deployment
metadata:
...
  generation: 3                    # there have been rollouts
  name: earth-3cc-web
  namespace: earth
...
spec:
...
  template:
    metadata:
      creationTimestamp: null
      labels:
        id: earth-3cc-web
    spec:
      containers:
      - image: nginx:1.16.1-alpine
        imagePullPolicy: IfNotPresent
        name: nginx
        readinessProbe:
          failureThreshold: 3
          initialDelaySeconds: 10
          periodSeconds: 20
          successThreshold: 1
          tcpSocket:
            port: 82               # this port doesn't seem to be right, should be 80
          timeoutSeconds: 1
...
```

We change the readiness-probe port, save and check the *Pods*:

```
➜ k -n earth get pod -l id=earth-3cc-web
NAME                              READY   STATUS    RESTARTS   AGE
earth-3cc-web-d49645966-52vb9     0/1     Running   0          6s
earth-3cc-web-d49645966-5tts6     0/1     Running   0          6s
earth-3cc-web-d49645966-db5gp     0/1     Running   0          6s
earth-3cc-web-d49645966-mk7gr     0/1     Running   0          6s
```

Running, but still not in ready state. Wait 10 seconds (initialDelaySeconds of readinessProbe) and check again:

```
➜ k -n earth get pod -l id=earth-3cc-web
NAME                              READY   STATUS    RESTARTS   AGE
earth-3cc-web-d49645966-52vb9     1/1     Running   0          32s
earth-3cc-web-d49645966-5tts6     1/1     Running   0          32s
earth-3cc-web-d49645966-db5gp     1/1     Running   0          32s
earth-3cc-web-d49645966-mk7gr     1/1     Running   0          32s
```

Let's check the service again:

```
➜ k run tmp --restart=Never --rm -i --image=nginx:alpine -- curl -m 5 earth-3cc-web.earth:6363
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   612  100   612    0     0  55636      0 --:--:-- --:--:-- --:--:-- 55636
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
    body {
        width: 35em;
        margin: 0 auto;
        font-family: Tahoma, Verdana, Arial, sans-serif;
    }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
...
```

We did it! Finally we write the reason into the requested location:

```
vim /opt/course/p3/ticket-654.txt
```

```
# /opt/course/p3/ticket-654.txt
yo Dirk, wrong port for readinessProbe defined!
```

CKAD Tips Kubernetes 1.33

In this section we'll provide some tips on how to handle the CKAD exam and browser terminal.

Knowledge

- Study all topics as proposed in the curriculum till you feel comfortable with all
- Learn and Study the in-browser scenarios on https://killercoda.com/killer-shell-ckad
- Read this and do all examples: https://kubernetes.io/docs/concepts/cluster-administration/logging
- Understand Rolling Update Deployment including maxSurge and maxUnavailable
- Do 1 or 2 test session with this CKAD Simulator. Understand the solutions and maybe try out other ways to achieve the same
- Be fast and breath `kubectl`

CKAD Preparation

Read the Curriculum

https://github.com/cncf/curriculum

Read the Handbook

https://docs.linuxfoundation.org/tc-docs/certification/lf-handbook2

Read the important tips

https://docs.linuxfoundation.org/tc-docs/certification/tips-cka-and-ckad

Read the FAQ

https://docs.linuxfoundation.org/tc-docs/certification/faq-cka-ckad

Kubernetes documentation

Get familiar with the Kubernetes documentation and be able to use the search. Allowed resources are:

- https://kubernetes.io/docs
- https://kubernetes.io/blog
- https://helm.sh/docs

ℹ️ Verify the list here

The Exam UI / Remote Desktop

The real exam, as well as the simulator, provides a Remote Desktop (XFCE) on Ubuntu/Debian. Coming from OSX/Windows there will be changes in copy&paste for example.

Official Information

ExamUI: Performance Based Exams

Lagging

There could be some lagging, definitely make sure you are using a good internet connection because your webcam and screen are transferring all the time.

Kubectl autocompletion and commands

The following are installed or pre-configured, verify the list here:

- `kubectl` with `k` alias and Bash autocompletion
- `yq` or YAML processing
- `curl` and `wget` for testing web services
- `man` and man pages for further documentation

ℹ️ You're allowed to install tools, like `tmux` for terminal multiplexing or `jq` for JSON processing

Copy & Paste

Copy and pasting will work like normal in a Linux Environment:

What always works: copy+paste using right mouse context menu What works in Terminal: Ctrl+Shift+c and Ctrl+Shift+v What works in other apps like Firefox: Ctrl+c and Ctrl+v

Score

There are 15-20 questions in the exam. Your results will be automatically checked according to the handbook. If you don't agree with the results you can request a review by contacting the Linux Foundation Support.

Notepad & Flagging Questions

You can flag questions to return to later. This is just a marker for yourself and won't affect scoring. You also have access to a simple notepad in the browser which can be used to store any kind of plain text. It might make sense to use this and write down additional information about flagged questions. Instead of using the notepad you could also open Mousepad (XFCE application inside the Remote Desktop) or create a file with Vim.

VSCodium

You can use VSCodium to edit files and you can also use its terminal to run commands. You're not allowed to install any VSCodium extensions.

Servers

Each question needs to be solved on a specific instance other than your main terminal. You'll need to connect to the correct instance via ssh, the command is provided before each question.

PSI Bridge

Starting with PSI Bridge:

- The exam will now be taken using the PSI Secure Browser, which can be downloaded using the newest versions of Microsoft Edge, Safari, Chrome, or Firefox
- Multiple monitors will no longer be permitted
- Use of personal bookmarks will no longer be permitted

The new ExamUI includes improved features such as:

- A remote desktop configured with the tools and software needed to complete the tasks
- A timer that displays the actual time remaining (in minutes) and provides an alert with 30, 15, or 5 minute remaining
- The content panel remains the same (presented on the Left Hand Side of the ExamUI)

Read more here.

Terminal Handling

Bash Aliases

In the real exam, each question has to be solved on a different instance to which you connect via ssh. This means it's not advised to configure bash aliases because they wouldn't be available on the instances accessed by ssh.

Be fast

Use the `history` command to reuse already entered commands or use even faster history search through **Ctrl r** .

If a command takes some time to execute, like sometimes `kubectl delete pod x`. You can put a task in the background using **Ctrl z** and pull it back into foreground running command `fg`.

You can delete *pods* fast with:

```
k delete pod x --grace-period 0 --force
```

Vim

Be great with vim.

Settings

In case you face a situation where vim is not configured properly and you face for example issues with pasting copied content you should be able to configure via `~/.vimrc` or by entering manually in vim settings mode:

```
set tabstop=2
set expandtab
set shiftwidth=2
```

The `expandtab` make sure to use spaces for tabs.

Note that changes in `~/.vimrc` will not be transferred when connecting to other instances via ssh.

Toggle vim line numbers

When in `vim` you can press **Esc** and type `:set number` or `:set nonumber` followed by **Enter** to toggle line numbers. This can be useful when finding syntax errors based on line - but can be bad when wanting to mark&copy by mouse. You can also just jump to a line number with **Esc** `:22` + **Enter**.

Copy&Paste

Get used to copy/paste/cut with vim:

```
Mark lines: Esc+V (then arrow keys)
Copy marked lines: y
Cut marked lines: d
Past lines: p or P
```

Indent multiple lines

To indent multiple lines press **Esc** and type `:set shiftwidth=2`. First mark multiple lines using `Shift v` and the up/down keys. Then to indent the marked lines press `>` or `<`. You can then press `.` to repeat the action.

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